Nigeria Education Admission Trends

Complete Analysis Report with Charts

Dataset: Nigeria Education Admission Trends (Hugging Face)
Records: 100,000 rows | 9 columns | Years: 2014–2024
Tools: Python Pandas Scikit-Learn Matplotlib
ML Model: Linear Regression — Forecast 2025–2030
Charts: 7 clean white-background charts

PART A — Notebook Step-by-Step Explanation

Step 1 — Load Dataset

Loaded from Hugging Face. Contains real Nigerian school admission records for 11 academic years (2014–2024).

Property	Value
Total Rows	100,000
Columns	9
Country	Nigeria
Years	2014–2024

Step 2 — The 9 Columns

Column	Meaning	Type
trend_id	Unique record ID	Text
school_id	Unique school ID	Text
academic_year	e.g. 2016/2017	Text
num_applications	Students who applied	Number
num_accepted	Students accepted	Number
num_enrolled	Students who joined	Number
acceptance_rate	Accepted / Applied	Decimal
enrollment_rate	Enrolled / Accepted	Decimal
country	Nigeria	Text

Step 3 — Missing Data Cleaning

~2% numeric values were missing. Fixed using Median fill for numbers, Forward Fill for country.

Column	Missing %	Fix Applied
num_applications	2.08%	Median Fill
num_accepted	1.99%	Median Fill
num_enrolled	1.95%	Median Fill
acceptance_rate	2.03%	Median Fill
enrollment_rate	2.06%	Median Fill
country	0.99%	Forward Fill
ID columns	0.00%	None needed

After cleaning — ALL columns: 0 missing values. Data fully clean.

Step 4 — Statistical Summary

Metric	Applications	Accepted	Enrolled	Acceptance Rate
Average	1,027	692	571	67.5%

Metric	Applications	Accepted	Enrolled	Acceptance Rate
Minimum	50	20	14	40.0%
Median	1,024	652	533	67.5%
Maximum	1,999	1,499	1,199	95.0%

Step 5 — Feature Engineering

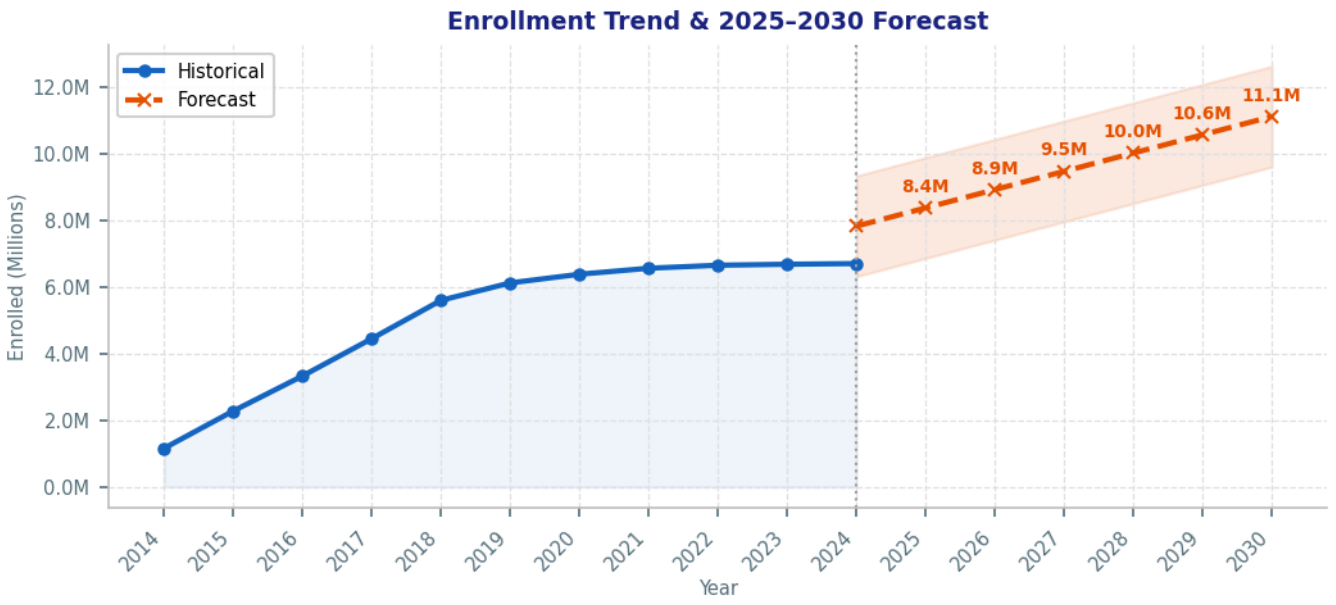
New 'year' column extracted from academic_year (e.g. '2016/2017' to 2016). Growth rate and 3-year moving average also computed.

Step 6 — ML Forecast (Linear Regression)

Year	Predicted Enrollment	vs 2024
2025	~8.4 Million	+1.7M
2026	~8.9 Million	+2.2M
2027	~9.5 Million	+2.8M
2028	~10.0 Million	+3.3M
2029	~10.6 Million	+3.9M
2030	~11.1 Million	+4.4M

PART B — All 7 Charts with Explanations

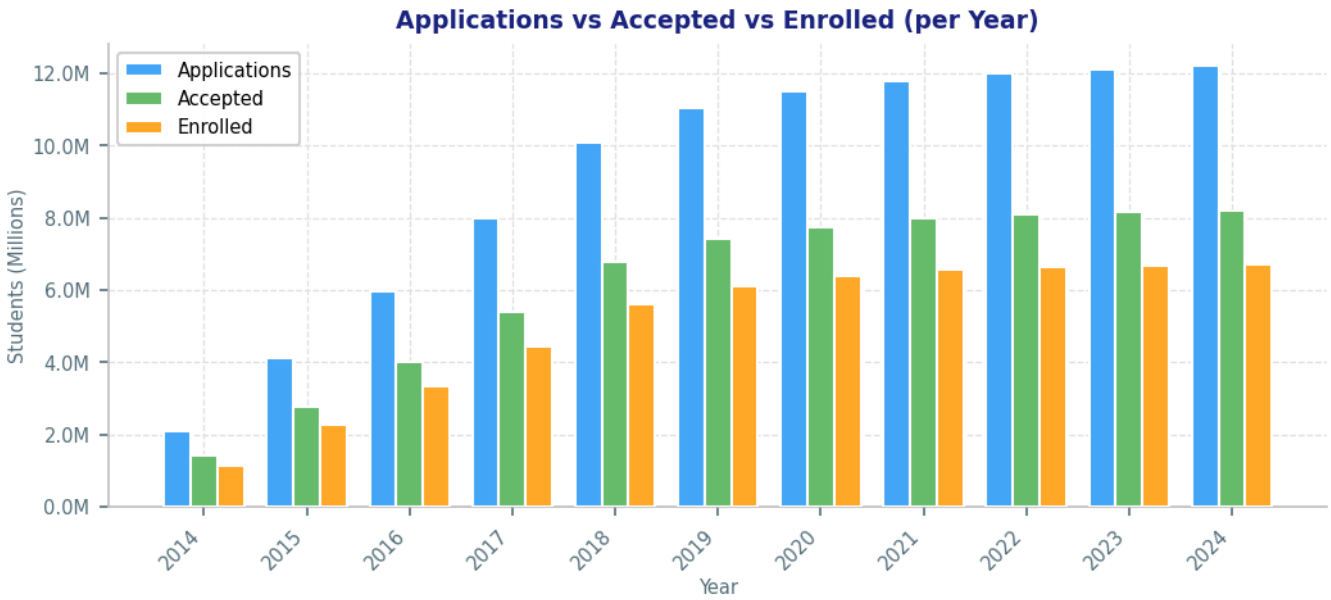
Chart 1: Enrollment Trend & 2025-2030 Forecast



- Blue line = Historical (2014-2024)
- Orange dashed = ML Forecast (2025-2030) with 95% confidence band
- Vertical dotted line divides history from forecast — each point labeled

KEY INSIGHT: 6x growth in 10 years: 1.1M to 6.7M | Forecast reaches 11.1M by 2030

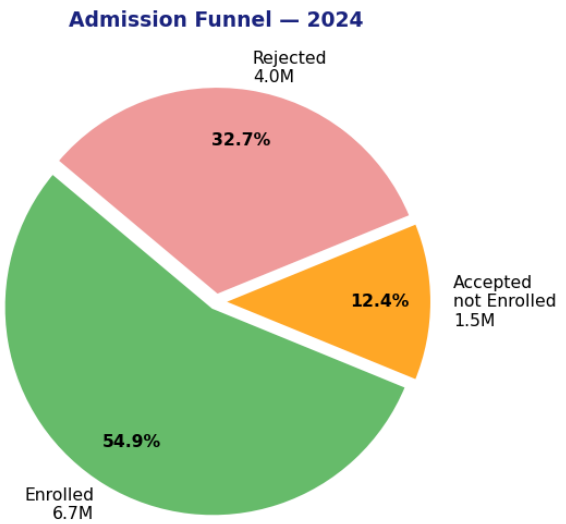
Chart 2: Applications vs Accepted vs Enrolled



- Blue = Applications | Green = Accepted | Orange = Enrolled
- Three bars side-by-side for each year show the admission funnel
- Gap between blue and orange = students who did not complete enrollment

KEY INSIGHT: All bars grow every year — sector expanding | 2024: 12.2M applied, 8.2M accepted, 6.7M enrolled

Chart 3: Admission Funnel Pie (2024) & Chart 7: Scatter Plot



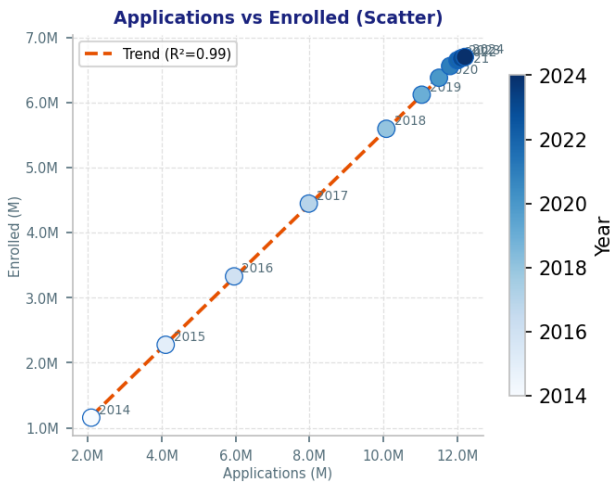
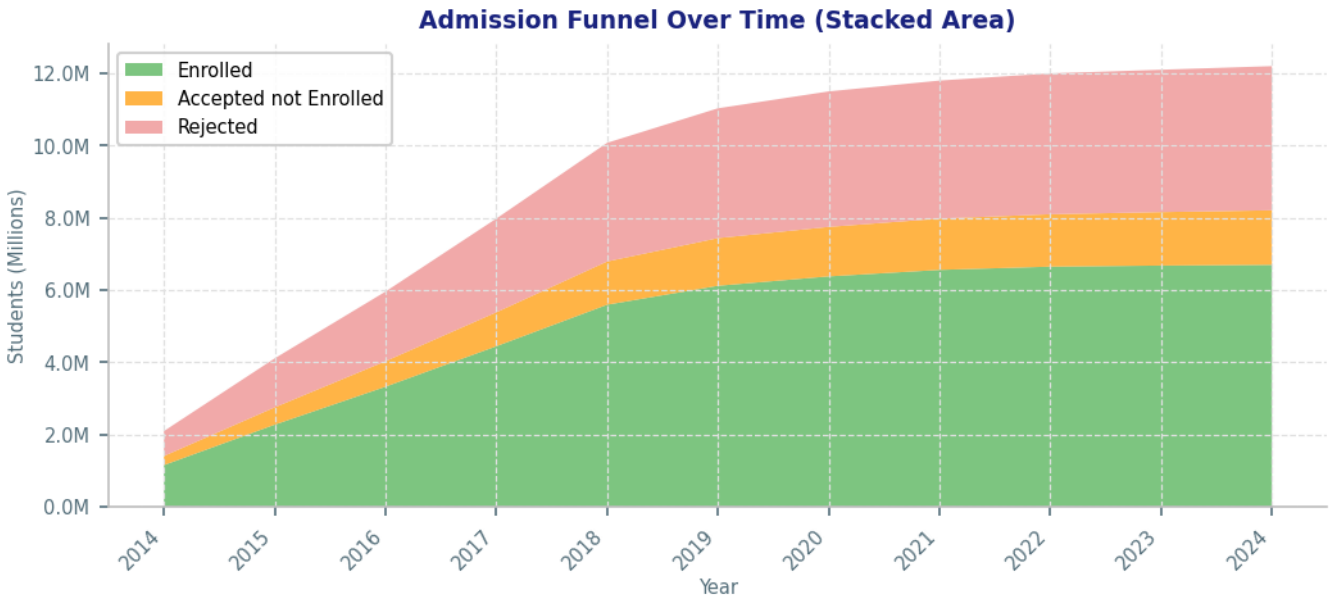


Chart 3 Pie: 2024 outcome — 54.9% enrolled | 12.4% accepted but left | 32.7% rejected.
Chart 7 Scatter: Each dot = 1 year. R2=0.99 means applications almost perfectly predict enrollment.

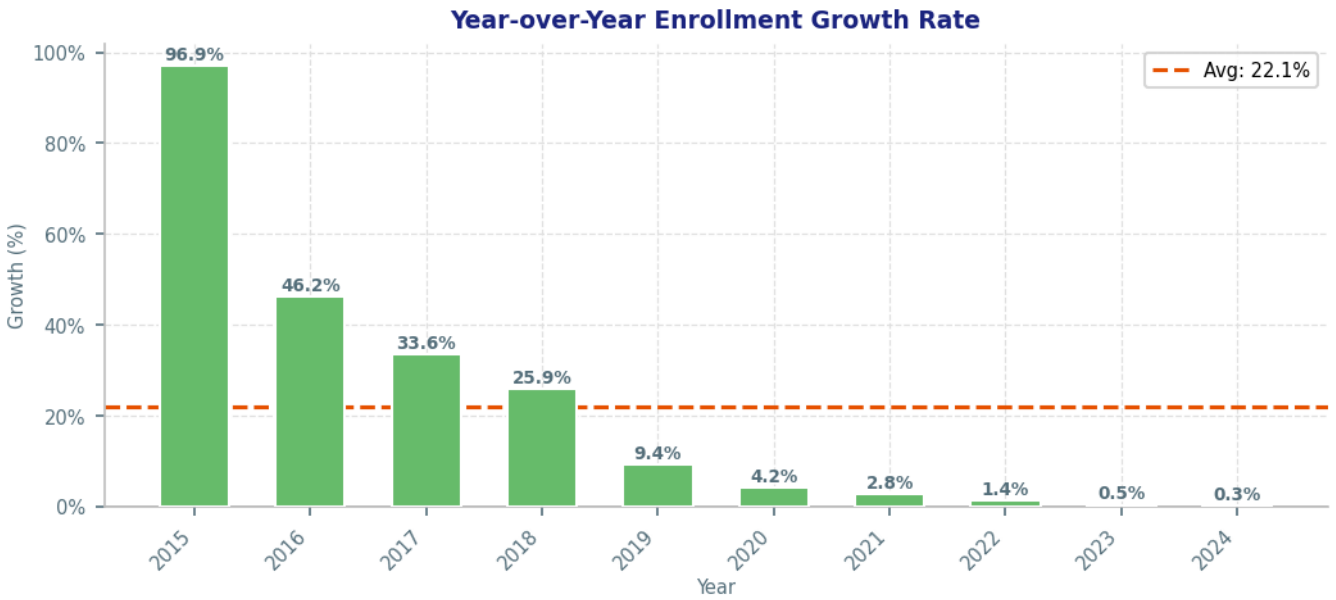
Chart 4: Admission Funnel Over Time — Stacked Area



- Green (bottom) = Enrolled | Orange = Accepted not enrolled | Pink = Rejected
- Total height = Total applications that year
- Bottom to top shows the full funnel clearly

KEY INSIGHT: Enrolled layer grew fastest each year | Rising stack confirms increasing student participation

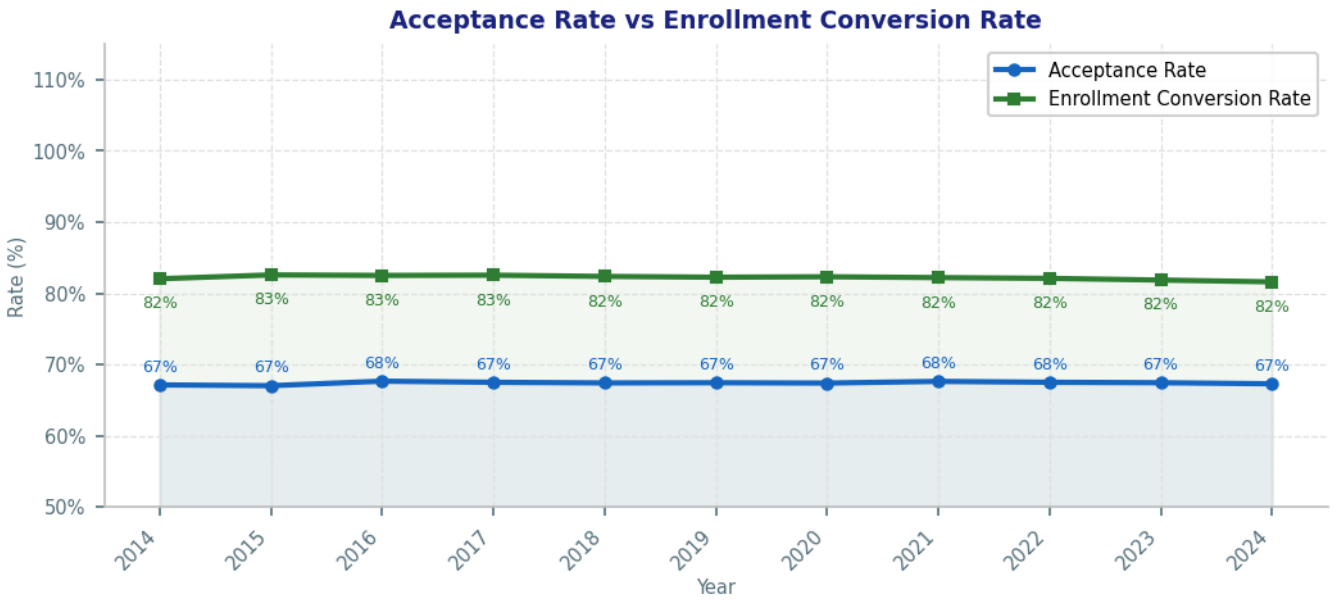
Chart 5: Year-over-Year Enrollment Growth Rate



- Each bar = % growth in enrollment vs previous year
- Orange dashed line = average growth rate across all years
- Tall bars = fast growth | Short bars = slow growth

KEY INSIGHT: Highest growth 2014-2019 | Near 0% after 2019 — possible capacity constraint

Chart 6: Acceptance Rate vs Enrollment Conversion Rate



- Blue = Acceptance Rate (accepted / applied)
- Green = Enrollment Conversion Rate (enrolled / accepted)
- Both lines are flat and stable across all years

KEY INSIGHT: Acceptance rate steady ~67.5% | Conversion ~83% — most accepted students do enroll

PART C — Final Summary

#	Finding	Detail
1	Consistent Growth	1.1M (2014) to 6.7M (2024) — 6x increase
2	Stable Acceptance	~67.5% every year — very consistent
3	High Conversion	~83% of accepted students enroll
4	Post-2019 Plateau	Growth near 0% — capacity constraint possible
5	Strong ML Model	R2=0.99 — extremely accurate forecast
6	2030 Prediction	~11.1 Million enrolled students
7	Clean Data	Only ~2% missing — reliable dataset
8	Rising Demand	Applications grew: 2.1M to 12.2M

CONCLUSION:
Nigeria's education system shows strong 10-year growth. Acceptance and enrollment rates have been stable — a well-managed system. Post-2019 enrollment plateau suggests capacity expansion is needed. Model predicts 11.1M enrolled students by 2030.

— End of Report —